GROUP 9 · SDG2410J · ROADSHOW CARD

Semester Footprint — the numbers, then the answers

Every figure re-checked in the browser on 10 November 2026. Nothing here is new; it is the same file you are about to open.

DEFAULT · 18 WEEKS

783.7
kg reported

96.1
sourced CO_2

687.6
unverified CO_2e

Transport 63.4 kg

Walk 360 km · 0.0 kg
Metro 540 km · 21.6 kg
Ride-hail 90 km · 18.0 kg
Rail 880 km · 23.8 kg

Food 648.0 kg

Beef 54 × 100 g · 324.0
Other meat 144 × 250 g · 216.0
Fish/egg/soy 90 × 400 g · 72.0
Plant 90 × 400 g · 36.0

Energy 72.3 kg

Room 120 kWh / 30 days
→ 504 kWh / 126 days
His quarter 126 kWh
× 0.5737 kg CO_2/kWh

THE ONE CHANGE · 16 WEEKS · THEN RESET

699.3
kg reported

Sourced 88.1 · unverified 611.2. Metro 480 km · beef 48 meals · electricity 112 days. Rail stays 880 km — it was already per semester.

THE TWO FACTORS WITH A SOURCE · BOTH REPORT CO_2, NOT CO_2e

Electricity 0.5737 kg CO_2/kWh · Shanghai provincial average, 2023 · Announcement 2025 No. 47

High-speed rail 0.02709 kg CO_2/passenger·km · China 2019, operation phase · Tong et al. 2025

THE TWO SUGGESTIONS TO READ ALOUD

Beef −159.3 → 624.4 "illustrative": the per-kilogram factor is provisional and the portion is our assumption.

Electricity −14.5 → 769.3 the factor has a source, so the arithmetic is sourced; the 20 percent is still a scenario.

IF ASKED

Why 0.5737 and not the class sheet's 0.4701?

That one is Shanghai 2024 and it is CO_2e. This one is 2023 provincial average, and the source reports CO_2.

Why not multiply by 52?

The profile mixes weeks, months, 30 days and semesters, and never says how many semesters make a year. One month = 4 weeks.

Why is food so large?

3 beef meals a week × 18 weeks × an assumed 100 g portion × a provisional 60 kg CO_2e/kg. The portion is ours, so the ranking is illustrative.

Why is your figure below 2.54 and 3.84?

Two 18-week semesters is about 1.6 t. Our food alone is 1.3 t, against 1.24 t in Li et al. 2015 over about the same nine months. Showers, study, goods and waste are outside.

Why leave out showers and waste?

The profile gives counts, not energy, mass, or a pathway. Li et al. put showers at 0.65 t a year, 18 per cent of their total, so the gap is material — we just cannot fill it honestly.

Which paper is your boundary closer to?

Zhu's on food, because our factors are per kilogram of food. Li's on electricity, because we count only the room meter. We sit between them, and lower, because goods and waste are out.

Is 120 kWh Li Ming's own use?

No. It is the whole room over 30 days. He is one quarter of four, scaled to the semester.

A battery car is always lower, right?

Only if asked. iX5 production 18.1 t against 11.7 t for the diesel X5; over 200,000 km on the European mix, 31.1 t against 58.3 t. Not an input to this calculator.